

# Supplementary Material: Exact Structural Abstraction and Tractability Limits

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## 1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

| ID   | Lean Handle / Source                                                              | ID   | Lean Handle / Source                                                                            |
|------|-----------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------|
| FR1  | currentFamilies_partition_by_role<br>Paper4dFrontier/Classification.lean          | FR2  | currentFamilies_partition_counts<br>Paper4dFrontier/Classification.lean                         |
| FR3  | current_families_have_finite_basis<br>Paper4dFrontier/Classification.lean         | FR4  | current_positive_interfaces_factor_through_role<br>Paper4dFrontier/Classification.lean          |
| FR5  | lifted_families_reduce_to_core<br>Paper4dFrontier/Classification.lean             | FR6  | degenerateFamilies_complete<br>Paper4dFrontier/Classification.lean                              |
| FR7  | exists_decisionProblem_realizing_labeling<br>Paper4dFrontier/Realizability.lean   | FR8  | realizingProblem_decisionEquiv_iff<br>Paper4dFrontier/Realizability.lean                        |
| FR13 | isOptimal_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean        | FR14 | opt_eq_positiveAffineTransform<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR15 | isSufficient_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean     | FR16 | isRelevant_positiveAffineTransform_iff<br>Paper4dFrontier/FamilyAxioms.lean                     |
| FR17 | decisionEquiv_relabelActions_iff<br>Paper4dFrontier/FamilyAxioms.lean             | FR18 | decisionEquiv_relabelStates_iff<br>Paper4dFrontier/FamilyAxioms.lean                            |
| FR19 | isSufficient_relabelActions_iff<br>Paper4dFrontier/FamilyAxioms.lean              | FR20 | isSufficient_relabelStates_iff<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR21 | allProblems_relabelInvariant<br>Paper4dFrontier/FamilyAxioms.lean                 | FR22 | allProblems_kernelUniversal<br>Paper4dFrontier/FamilyAxioms.lean                                |
| FR23 | relabelInvariant_not_enough<br>Paper4dFrontier/FamilyAxioms.lean                  | FR24 | optRangeCard_realizingIdentity<br>Paper4dFrontier/FamilyAxioms.lean                             |
| FR25 | currentFamilies_explicit<br>Paper4dFrontier/Classification.lean                   | FR26 | coreFamilies_explicit<br>Paper4dFrontier/Classification.lean                                    |
| FR27 | liftedFamilies_explicit<br>Paper4dFrontier/Classification.lean                    | FR28 | degenerateFamilies_explicit<br>Paper4dFrontier/Classification.lean                              |
| FR29 | primitiveMechanisms_explicit<br>Paper4dFrontier/Classification.lean               | FR30 | isSufficient_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                      |
| FR31 | isRelevant_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean          | FR32 | isIrrelevant_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                      |
| FR33 | isMinimalSufficient_iff_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean | FR34 | sufficientSets_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                     |
| FR35 | relevantSet_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean          | FR37 | quotientCard_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                       |
| FR38 | quotientCard_realizingIdentity<br>Paper4dFrontier/FamilyAxioms.lean               | FR39 | realizingProblemQuotientEquivLabelRange_apply_quotientMap<br>Paper4dFrontier/Realizability.lean |

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| ID    | Lean Handle / Source                                                                                              | ID    | Lean Handle / Source                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------|
| FR40  | realizingProblem_quotientCard_eq_labelRangeCard<br>Paper4dFrontier/Realizability.lean                             | FR41  | certificationStatistic_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                              |
| FR42  | sufficientSetCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                                   | FR43  | minimalSufficientSetCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                           |
| FR44  | relevantCoordCount_eq_of_decisionEquiv_iff<br>Paper4dFrontier/FamilyAxioms.lean                                   | FR45  | setoidRealizingProblem_decisionEquiv_iff<br>Paper4dFrontier/Realizability.lean                                   |
| FR46  | setoidRealizingProblemQuotientEquivSetoidQuotient_apply_quotientMap<br>Paper4dFrontier/Realizability.lean         | FR47  | exists_decisionProblem_realizing_setoid<br>Paper4dFrontier/Realizability.lean                                    |
| FR48  | isSufficient_iff_agreementRel_le_decisionEquiv<br>Paper4dFrontier/FamilyAxioms.lean                               | FR49  | isIrrelevant_iff_sufficient_erase<br>Paper4dFrontier/FamilyAxioms.lean                                           |
| FR50  | isRelevant_iff_not_sufficient_erase<br>Paper4dFrontier/FamilyAxioms.lean                                          | FR51  | quotientEquiv_of_decisionEquiv_iff_apply_quotientMap<br>Paper4dFrontier/FamilyAxioms.lean                        |
| FR52  | empty_sufficient_of_constant_opt<br>Paper4dFrontier/ClosureLaws.lean                                              | FR53  | irrelevant_of_constant_opt<br>Paper4dFrontier/ClosureLaws.lean                                                   |
| FR54  | decisionEquiv_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                             | FR55  | isSufficient_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR56  | isRelevant_duplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                                | FR57  | decisionEquiv_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR58  | isSufficient_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                               | FR59  | isRelevant_duplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                                |
| FR60  | duplicateStateQuotientEquivOriginal_apply_quotientMap<br>Paper4dFrontier/ClosureLaws.lean                         | FR66  | booleanCube_card<br>Paper4dFrontier/EnumerationBounds.lean                                                       |
| FR67  | exists_booleanCube_larger_than_monomial<br>Paper4dFrontier/EnumerationBounds.lean                                 | FR68  | decisionEquiv_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                          |
| FR69  | isSufficient_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                            | FR70  | isRelevant_addDuplicateState_iff<br>Paper4dFrontier/ClosureLaws.lean                                             |
| FR71  | addDuplicateStateQuotientEquivOriginal_apply_quotientMap<br>Paper4dFrontier/ClosureLaws.lean                      | FR72  | some_mem_opt_addDuplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                          |
| FR73  | none_mem_opt_addDuplicateAction_iff<br>Paper4dFrontier/ClosureLaws.lean                                           | FR83  | isSufficient_noiseExtended_iff<br>Paper4dFrontier/DimensionalNoiseExtension.lean                                 |
| FR84  | isRelevant_noiseExtended_iff<br>Paper4dFrontier/DimensionalNoiseExtension.lean                                    | FR85  | lastCoord_irrelevant_noiseExtended<br>Paper4dFrontier/DimensionalNoiseExtension.lean                             |
| FR100 | parentTree_realTreewidth_one<br>Paper4dFrontier/ParentTreewidth.lean                                              | FR101 | low_rank_only_witness<br>Paper4dFrontier/LowRankOnlyWitness.lean                                                 |
| FR105 | separable_only_witness<br>Paper4dFrontier/SeparableOnlyWitness.lean                                               | FR106 | bounded_actions_only_witness<br>Paper4dFrontier/BoundedActionsOnlyWitness2.lean                                  |
| FR108 | all_independent_core_mechanisms_nondegenerate<br>Paper4dFrontier/Block2Progress.lean                              | FR110 | parentTreeStructured_implies_treeStructured<br>Paper4dFrontier/ParentTreewidth.lean                              |
| FR111 | treeStructured_and_unique_parent_iff_parentTreeStructured<br>Paper4dFrontier/ParentTreewidth.lean                 | FR112 | bounded_treewidth_only_witness<br>Paper4dFrontier/BoundedTreewidthOnlyWitness.lean                               |
| FR113 | symmetry_only_witness<br>Paper4dFrontier/SymmetryOnlyWitness.lean                                                 | FR114 | weak_tree_structured_not_width_one<br>Paper4dFrontier/LegacyTreeGap.lean                                         |
| FR115 | tree_structure_hardness_bridge<br>Paper4dFrontier/TreeStructureHardnessBridge.lean                                | FR118 | pairCrossDifference_eq_binaryCrossDifference_of_lt<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean               |
| FR119 | pairwise_zero_crossDifference_unaryDecomposition<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean                  | FR120 | binary_pairwise_symmetry_dichotomy<br>Paper4dFrontier/BinaryPairwiseDichotomy.lean                               |
| FR121 | actionGapCrossDifference_eq_binaryCrossDifference_of_lt<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean | FR122 | decisionRelevant_zero_actionGap_implies_unaryReduction<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean |
| FR123 | binary_pairwise_symmetry_decision_relevant_dichotomy<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean    | FR124 | block6_genuine_interaction_dichotomy_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                       |

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| ID    | Lean Handle / Source                                                                                                                | ID    | Lean Handle / Source                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| FR125 | decisionRelevantInteractionGraph_<br>addActionOffset<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean                      | FR126 | action_offset_can_force_constant_optimizer<br>Paper4dFrontier/Block6Obstruction.lean                                                 |
| FR127 | block6_action_offset_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                          | FR128 | offsetNormalizedDecisionRelevantInteractionGraph_<br>wellDefined<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean           |
| FR129 | block6_offset_normalized_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                      | FR130 | neverOptimalGhost_decisionRelevantGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                             |
| FR131 | neverOptimalGhost_supportedGraph_eq_bot<br>Paper4dFrontier/Block6Obstruction.lean                                                   | FR132 | support_filtering_removes_known_block6_<br>obstructions<br>Paper4dFrontier/Block6Obstruction.lean                                    |
| FR133 | block6_ghost_action_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                           | FR134 | marginMasking_supportedGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                                        |
| FR135 | marginMasking_zero_sufficient<br>Paper4dFrontier/Block6Obstruction.lean                                                             | FR136 | block6_optimizer_supported_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                     |
| FR137 | MarginBounded<br>Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean                                                             | FR138 | dominantPair_marginBounded<br>Paper4dFrontier/Block6Obstruction.lean                                                                 |
| FR139 | dominantPair_supportedGraph_eq_top<br>Paper4dFrontier/Block6Obstruction.lean                                                        | FR140 | block6_margin_bounded_obstruction<br>Paper4dFrontier/Block6Obstruction.lean                                                          |
| FR145 | not_collapseLandscapeInfinity_of_oracle_<br>predicate<br>Paper4dFrontier/MetaCharacterization.lean                                  | FR176 | DecisionQuotient.DecisionProblem.quotient_is_<br>coarsest<br>paper4/DecisionQuotient/Quotient.lean                                   |
| FR177 | DecisionQuotient.DecisionProblem.quotient_has_<br>unique_factorization<br>paper4/DecisionQuotient/Quotient.lean                     | FR178 | DecisionQuotient.DecisionProblem.srank_eq_<br>relevant_card<br>paper4/DecisionQuotient/Tractability/<br>StructuralRank.lean          |
| FR179 | DecisionQuotient.quotientEntropy_le_srank_<br>binary<br>paper4/DecisionQuotient/Information.lean                                    | FR180 | DecisionQuotient.Statistics.fisherMatrix_rank_<br>eq_srank<br>paper4/DecisionQuotient/Statistics/<br>FisherInformation.lean          |
| FR181 | DecisionQuotient.Information.compression_<br>below_srank_fails<br>paper4/DecisionQuotient/Information/<br>RDSrank.lean              | FR182 | DecisionQuotient.Information.srank_bits_<br>sufficient<br>paper4/DecisionQuotient/Information/<br>RDSrank.lean                       |
| FR183 | DecisionQuotient.Physics.single_future_zero_<br>cost<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean               | FR184 | DecisionQuotient.Physics.transportCost_pos_of_<br>offDiag<br>paper4/DecisionQuotient/Physics/<br>WassersteinIntegrity.lean           |
| FR185 | no_closureInvariant_predicate_of_orbit_gap<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                   | FR186 | no_admissibleNormalizationPredicate_decides_<br>dominantPair<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                  |
| FR187 | no_admissibleNormalizationPredicate_decides_<br>marginBounded<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                | FR188 | no_admissibleNormalizationPredicate_decides_<br>ghostActionTwoPairCrossOne<br>Paper4dFrontier/ObstructionPredicateCandidates.lean    |
| FR189 | no_admissibleNormalizationPredicate_decides_<br>offsetActionZeroPairCrossOne<br>Paper4dFrontier/ObstructionPredicateCandidates.lean | FR190 | admissibleCollapseLandscapeInfinity_full_paper<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                |
| FR191 | ClosureSoundPackage<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                                          | FR192 | ReasonableGuardrailPackage<br>Paper4dFrontier/ObstructionPredicateCandidates.lean                                                    |
| FR193 | no_closureSoundPackage_predicate_decides_four_<br>obstruction_families<br>Paper4dFrontier/ObstructionPredicateCandidates.lean       | FR194 | no_reasonableGuardrailPackage_predicate_<br>decides_four_obstruction_families<br>Paper4dFrontier/ObstructionPredicateCandidates.lean |
| FR197 | classifier_agrees_on_closureEquivalent_of_<br>correctOnDomain<br>Paper4dFrontier/AdmissibleCharacterization.lean                    | FR198 | no_correctOnDomain_classifier_of_orbit_gap<br>Paper4dFrontier/AdmissibleCharacterization.lean                                        |
| FR199 | correct_classifier_inherits_<br>closureLawInvariant<br>Paper4dFrontier/AdmissibleCharacterization.lean                              | FR200 | admissibleNormalizationPredicate_has_explicit_<br>inhabitants<br>Paper4dFrontier/AdmissibleCharacterization.lean                     |
| FR201 | closureLawInvariant_of_iff_of_<br>closureEquivalent<br>Paper4dFrontier/AdmissibleCharacterization.lean                              | FR202 | exists_orbit_gap_of_not_closureLawInvariant<br>Paper4dFrontier/AdmissibleCharacterization.lean                                       |

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| ID    | Lean Handle / Source                                                                                                   | ID    | Lean Handle / Source                                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------|
| FR203 | closureLawInvariant_iff_no_orbit_gap<br>Paper4dFrontier/AdmissibleCharacterization.lean                                | FR204 | no_exact_closureLawInvariant_classifier_iff_exists_orbit_gap<br>Paper4dFrontier/AdmissibleCharacterization.lean                         |
| FR205 | distinctActionCount<br>Paper4dFrontier/DistinctActionProfiles.lean                                                     | FR206 | actionCount_profileCompressedSlice<br>Paper4dFrontier/DistinctActionProfiles.lean                                                       |
| FR207 | decisionEquiv_profileCompressedSlice_iff<br>Paper4dFrontier/DistinctActionProfiles.lean                                | FR208 | profileCompressedSlice_preserves_exactCertification<br>Paper4dFrontier/DistinctActionProfiles.lean                                      |
| FR209 | profileCompressedSlice_bounded_actions<br>Paper4dFrontier/DistinctActionProfiles.lean                                  | FR210 | OrbitGapOn<br>Paper4dFrontier/AdmissibleCharacterization.lean                                                                           |
| FR211 | no_orbitGapOn_of_exact_classifiable_by_closureLawInvariant_onDomain<br>Paper4dFrontier/AdmissibleCharacterization.lean | FR212 | exact_classifiable_by_closureLawInvariant_onDomain_iff_no_orbitGapOn<br>Paper4dFrontier/AdmissibleCharacterization.lean                 |
| FR213 | no_exact_closureLawInvariant_classifier_onDomain_iff_orbitGapOn<br>Paper4dFrontier/AdmissibleCharacterization.lean     | FR214 | boundedPatternScheme_holds_largeActionCount_iff<br>Paper4dFrontier/AdmissibleCharacterization.lean                                      |
| FR215 | boundedPatternDefinable_eventually_constant_in_actionCount<br>Paper4dFrontier/AdmissibleCharacterization.lean          | FR216 | boundedPatternDefinable_largeActionCount_agrees<br>Paper4dFrontier/AdmissibleCharacterization.lean                                      |
| FR217 | payloadSufficient_iff_realizingProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                              | FR218 | payloadRelevant_iff_realizingProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                                                   |
| FR219 | payloadIrrelevant_iff_realizingProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                              | FR220 | payloadMinimalSufficient_iff_realizingProblem_isMinimalSufficient<br>Paper4dFrontier/Realizability.lean                                 |
| FR221 | payloadSufficientSets_eq_realizingProblem_sufficientSets<br>Paper4dFrontier/Realizability.lean                         | FR222 | payloadRelevantSet_eq_realizingProblem_relevantSet<br>Paper4dFrontier/Realizability.lean                                                |
| FR223 | feasiblePayloadSufficient_iff_lawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                    | FR224 | feasiblePayloadRelevant_iff_lawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                                         |
| FR225 | feasiblePayloadIrrelevant_iff_lawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                    | FR226 | setValuedPayloadSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                           |
| FR227 | setValuedPayloadRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean              | FR228 | setValuedPayloadIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                           |
| FR229 | outputSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean           | FR230 | outputSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                                |
| FR231 | outputSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean           | FR232 | approximationSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                     |
| FR233 | approximationSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean        | FR234 | approximationSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                     |
| FR235 | outputSemanticsSufficientSets_eq_of_outputSemanticsEquivalent<br>Paper4dFrontier/Realizability.lean                    | FR236 | outputSemanticsRelevantSet_eq_of_outputSemanticsEquivalent<br>Paper4dFrontier/Realizability.lean                                        |
| FR237 | quotientMap_realizes_setoid_asOutputSemantics<br>Paper4dFrontier/Realizability.lean                                    | FR240 | exists_outputSemantics_realizing_setoid<br>Paper4dFrontier/Realizability.lean                                                           |
| FR241 | outputSemanticsSetoid<br>Paper4dFrontier/Realizability.lean                                                            | FR242 | SemanticallyExtensionalMap<br>Paper4dFrontier/Realizability.lean                                                                        |
| FR243 | semanticallyExtensionalMap_factors_through_outputSemanticsQuotient<br>Paper4dFrontier/Realizability.lean               | FR244 | semanticallyExtensionalClaim_factors_through_outputSemanticsQuotient<br>Paper4dFrontier/Realizability.lean                              |
| FR245 | optimizerComputation_polytime_closureLawInvariant<br>Paper4dFrontier/ComputeCostApplications.lean                      | FR246 | optimizerSetPayload_polytime_closureLawInvariant<br>Paper4dFrontier/ComputeCostApplications.lean                                        |
| FR247 | optimizerSetSearch_polytime_closureLawInvariant<br>Paper4dFrontier/ComputeCostApplications.lean                        | FR248 | optimizerComputation_polytime_classifier_agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostApplications.lean |

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| ID    | Lean Handle / Source                                                                                                                   | ID    | Lean Handle / Source                                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| FR249 | optimizerComputation_polytime_classifier_agrees_on_closureEquivalent<br>Paper4dFrontier/ComputeCostApplications.lean                   | FR250 | no_correct_optimizerComputation_polytime_classifier_decides_dominantPair<br>Paper4dFrontier/ComputeCostApplications.lean                            |
| FR251 | no_admissibleNormalizationPredicate_optimizerComputation_polytime_and_dominantPair<br>Paper4dFrontier/ComputeCostApplications.lean     | FR252 | CorrectnessForcesOrbitAgreementOnDomain<br>Paper4dFrontier/AdmissibleCharacterization.lean                                                          |
| FR253 | no_orbitGapOn_of_correct_classifier_onDomain_of_forcedOrbitAgreement<br>Paper4dFrontier/AdmissibleCharacterization.lean                | FR254 | correct_classifier_onDomain_iff_no_orbitGapOn_of_forcedOrbitAgreement<br>Paper4dFrontier/AdmissibleCharacterization.lean                            |
| FR255 | no_correct_classifier_onDomain_iff_orbitGapOn_of_forcedOrbitAgreement<br>Paper4dFrontier/AdmissibleCharacterization.lean               | FR256 | optimizerComputation_polytime_correct_classifier_onDomain_iff_no_orbitGapOn<br>Paper4dFrontier/ComputeCostApplications.lean                         |
| FR259 | optimizerSetPayload_polytime_classifier_agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostApplications.lean | FR260 | optimizerSetPayload_polytime_classifier_agrees_on_closureEquivalent<br>Paper4dFrontier/ComputeCostApplications.lean                                 |
| FR261 | optimizerSetSearch_polytime_classifier_agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostApplications.lean  | FR262 | optimizerSetSearch_polytime_classifier_agrees_on_closureEquivalent<br>Paper4dFrontier/ComputeCostApplications.lean                                  |
| FR263 | relevant_of_uniformApprox_of_strict_gap_witness<br>Paper4dFrontier/ApproximateAdmissibility.lean                                       | FR264 | relevance_can_flip_under_arbitrarily_small_uniform_perturbation<br>Paper4dFrontier/ApproximateAdmissibility.lean                                    |
| FR266 | not_sufficient_of_uniformApprox_of_strict_gap_witness<br>Paper4dFrontier/ApproximateAdmissibility.lean                                 | FR267 | sufficiency_can_flip_under_arbitrarily_small_uniform_perturbation<br>Paper4dFrontier/ApproximateAdmissibility.lean                                  |
| FR268 | pacGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                     | FR269 | pacGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                                      |
| FR270 | pacGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                     | FR271 | regretGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                               |
| FR272 | regretGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                      | FR273 | regretGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                               |
| FR274 | statisticalRiskSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                  | FR275 | statisticalRiskSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                                   |
| FR276 | statisticalRiskSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                  | FR277 | anytimeGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                              |
| FR278 | anytimeGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                     | FR279 | anytimeGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                              |
| FR280 | finiteHorizonGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean           | FR281 | finiteHorizonGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                            |
| FR282 | finiteHorizonGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean           | FR283 | statisticalGuaranteeSemanticsSufficientSets_eq_of_outputSemanticsEquivalent<br>Paper4dFrontier/Realizability.lean                                   |
| FR284 | statisticalGuaranteeSemanticsRelevantSet_eq_of_outputSemanticsEquivalent<br>Paper4dFrontier/Realizability.lean                         | FR285 | statisticalGuarantee_classifier_agrees_on_closureEquivalent_of_correctOnDomain_of_transfer<br>Paper4dFrontier/StatisticalSemanticsApplications.lean |
| FR286 | no_correctOnDomain_statisticalGuarantee_classifier_of_orbit_gap_of_transfer<br>Paper4dFrontier/StatisticalSemanticsApplications.lean   | FR287 | statisticalGuarantee_correct_classifier_onDomain_iff_no_orbitGapOn_of_transfer<br>Paper4dFrontier/StatisticalSemanticsApplications.lean             |
| FR288 | opt_eq_of_uniformApprox_of_uniformStrictGapCover<br>Paper4dFrontier/ApproximateAdmissibility.lean                                      | FR289 | sufficientSets_eq_of_uniformApprox_of_uniformStrictGapCover<br>Paper4dFrontier/ApproximateAdmissibility.lean                                        |
| FR290 | relevantSet_eq_of_uniformApprox_of_uniformStrictGapCover<br>Paper4dFrontier/ApproximateAdmissibility.lean                              | FR291 | randomizedGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient<br>Paper4dFrontier/Realizability.lean                           |

(continued...)

| ID    | Lean Handle / Source                                                                                                                                           | ID    | Lean Handle / Source                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| FR292 | randomizedGuaranteeSemanticsRelevant_iff_<br>totalizedLawDecisionProblem_isRelevant<br>Paper4dFrontier/Realizability.lean                                      | FR293 | randomizedGuaranteeSemanticsIrrelevant_iff_<br>totalizedLawDecisionProblem_isIrrelevant<br>Paper4dFrontier/Realizability.lean                     |
| FR294 | randomizedGuarantee_classifier_agrees_on_<br>closureEquivalent_of_correctOnDomain_of_<br>transfer<br>Paper4dFrontier/StatisticalSemanticsApplications.lean     | FR295 | no_correctOnDomain_randomizedGuarantee_<br>classifier_of_orbit_gap_of_transfer<br>Paper4dFrontier/StatisticalSemanticsApplications.lean           |
| FR296 | randomizedGuarantee_correct_classifier_<br>onDomain_iff_no_orbitGapOn_of_transfer<br>Paper4dFrontier/StatisticalSemanticsApplications.lean                     | FR297 | transferredSemantics_classifier_agrees_on_<br>closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/StatisticalSemanticsApplications.lean       |
| FR298 | no_correctOnDomain_transferredSemantics_<br>classifier_of_orbit_gap<br>Paper4dFrontier/StatisticalSemanticsApplications.lean                                   | FR299 | transferredSemantics_correct_classifier_<br>onDomain_iff_no_orbitGapOn<br>Paper4dFrontier/StatisticalSemanticsApplications.lean                   |
| FR300 | decisionEquiv_iff_of_uniformApprox_of_<br>uniformStrictGapCover<br>Paper4dFrontier/ApproximateAdmissibility.lean                                               | FR301 | isMinimalSufficient_iff_of_uniformApprox_of_<br>uniformStrictGapCover<br>Paper4dFrontier/ApproximateAdmissibility.lean                            |
| FR302 | outputSemanticsExactRelevanceProfile_eq_of_<br>outputSemanticsEquivalent<br>Paper4dFrontier/Realizability.lean                                                 | FR303 | outputSemanticsExactRelevanceProfile_eq_<br>totalizedLawDecisionProblem<br>Paper4dFrontier/Realizability.lean                                     |
| FR304 | agreeOn_univ_iff_eq_of_<br>finiteIndicatorCoordinateSpace<br>Paper4dFrontier/Realizability.lean                                                                | FR305 | finite_outputSemantics_<br>allCoordinatesSufficient<br>Paper4dFrontier/Realizability.lean                                                         |
| FR306 | finite_outputSemantics_realized_by_<br>exactCertification<br>Paper4dFrontier/Realizability.lean                                                                | FR307 | agreeOn_univ_iff_eq_of_<br>singletonIdentityCoordinateSpace<br>Paper4dFrontier/Realizability.lean                                                 |
| FR308 | outputSemantics_admits_coordinatePresentation<br>Paper4dFrontier/Realizability.lean                                                                            | FR311 | encodable_stateSpace_admits_<br>countableBooleanPresentation<br>Paper4dFrontier/Realizability.lean                                                |
| FR312 | encodable_discreteMeasurable_stateSpace_<br>admits_measurable_countableBooleanPresentation<br>Paper4dFrontier/Realizability.lean                               | FR313 | encodable_discreteTopological_stateSpace_<br>admits_continuous_countableBooleanPresentation<br>Paper4dFrontier/Realizability.lean                 |
| FR314 | agreeOn_univ_iff_eq_of_<br>finiteBinaryCoordinateSpace<br>Paper4dFrontier/Realizability.lean                                                                   | FR315 | finite_exactSpecification_admits_<br>lowDimBooleanPresentation<br>Paper4dFrontier/Realizability.lean                                              |
| FR316 | IdentityOutputClosureSpec.classifier_agrees_<br>on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                     | FR317 | IdentityOutputClosureSpec.no_correctOnDomain_<br>classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                       |
| FR318 | hypothesisOutput_classifier_agrees_on_<br>closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                              | FR319 | estimatorOutput_classifier_agrees_on_<br>closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                  |
| FR320 | policyOutput_classifier_agrees_on_<br>closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                                  | FR321 | randomizedProcedure_classifier_agrees_on_<br>closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean              |
| FR322 | no_correctOnDomain_hypothesisOutput_<br>classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                                             | FR323 | no_correctOnDomain_estimatorOutput_classifier_<br>of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                                 |
| FR324 | no_correctOnDomain_policyOutput_classifier_of_<br>orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                                                 | FR325 | no_correctOnDomain_randomizedProcedure_<br>classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                             |
| FR326 | TransportedOutputClosureSpec.classifier_<br>agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                  | FR327 | TransportedOutputClosureSpec.no_<br>correctOnDomain_classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                    |
| FR328 | IdentityOutputClosureSpec.correctOnDomain_iff_<br>correctOnDomain_transport<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                                 | FR329 | representationRelativeHypothesis_classifier_<br>agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean |
| FR330 | representationRelativeEstimator_classifier_<br>agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean               | FR331 | representationRelativePolicy_classifier_<br>agrees_on_closureEquivalent_of_correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean     |
| FR332 | representationRelativeRandomizedProcedure_<br>classifier_agrees_on_closureEquivalent_of_<br>correctOnDomain<br>Paper4dFrontier/ComputeCostExternalOutputs.lean | FR333 | no_correctOnDomain_representationRelativeHypothesis_<br>classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean                |

(continued...)

| ID    | Lean Handle / Source                                                                                                                    | ID    | Lean Handle / Source                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------|
| FR334 | no_correctOnDomain_representationRelativeEstimator_classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean           | FR335 | no_correctOnDomain_representationRelativePolicy_classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean     |
| FR336 | no_correctOnDomain_representationRelativeRandomizedProcedure_classifier_of_orbit_gap<br>Paper4dFrontier/ComputeCostExternalOutputs.lean | FR340 | no_leanPayloadTransfer<br>Paper4dFrontier/Realizability.lean                                                                   |
| FR341 | predicateTransfer<br>Paper4dFrontier/Realizability.lean                                                                                 | FR342 | sufficiency_is_relation_refinement<br>Paper4dFrontier/Realizability.lean                                                       |
| FR343 | relevance_is_erased_failure_of_refinement<br>Paper4dFrontier/Realizability.lean                                                         | FR345 | exactSemanticsQuotient_universal_characterization<br>Paper4dFrontier/Realizability.lean                                        |
| FR346 | exactnessMeansExactAgreementWithValidity<br>Paper4dFrontier/Realizability.lean                                                          | FR347 | closure_sound_package_no_go_uses_only_closureLawInvariant<br>Paper4dFrontier/FullBinaryPairwiseDomain.lean                     |
| FR348 | fullBinaryPairwiseDomain_closure_closed<br>Paper4dFrontier/FullBinaryPairwiseDomain.lean                                                | FR349 | fullBinaryPairwiseDomain_contains_four_obstruction_families<br>Paper4dFrontier/FullBinaryPairwiseDomain.lean                   |
| FR350 | no_correct_tractability_classifier_on_fullBinaryPairwiseDomain<br>Paper4dFrontier/FullBinaryPairwiseDomain.lean                         | FR351 | ExactCorrectnessSpecification.universal_scope_over_rigorously_specified_problems<br>Paper4dFrontier/Realizability.lean         |
| FR352 | DecisionQuotient.DecisionProblem.minimalSufficient_eq_relevant'<br>paper4/DecisionQuotient/Sufficiency.lean                             | FR353 | DecisionQuotient.DecisionProblem.sufficientSets_principal'<br>paper4/DecisionQuotient/Sufficiency.lean                         |
| FR354 | DecisionQuotient.srank_le_sufficient_card<br>paper4/DecisionQuotient/Tractability/StructuralRank.lean                                   | FR355 | closureLawInvariant_iff_closureHull_eq_self<br>Paper4dFrontier/AdmissibleCharacterization.lean                                 |
| FR356 | no_orbitGapOn_iff_closureHull_disjoint<br>Paper4dFrontier/AdmissibleCharacterization.lean                                               | FR357 | exact_classifiable_by_closureLawInvariant_onDomain_iff_closureHull_disjoint<br>Paper4dFrontier/AdmissibleCharacterization.lean |
| FR358 | closureHull_least_exact_classifier_onDomain_of_no_orbitGapOn<br>Paper4dFrontier/AdmissibleCharacterization.lean                         |       |                                                                                                                                |

## 2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

| Paper claim                                                                              | Lean handle                                                                               |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Corollary 2.48: Exact Approximation Semantics Transfer                                   | FR234, FR233, FR232                                                                       |
| Corollary 2.29: Boolean Payload Transfer                                                 | FR340                                                                                     |
| Corollary 2.9: Constant-Optimizer Collapse Forces Trivial Certification                  | FR52, FR53                                                                                |
| Corollary 6.21: Domain Restriction Helps Only by Removing Orbit Gaps                     | FR252, FR254, FR255, FR253, FR256                                                         |
| Corollary 2.54: Every State Equivalence Relation Is Realizable as Exact Output Semantics | FR240, FR237                                                                              |
| Corollary 2.23: Compute-Cost Version: External Output Objects                            | FR316, FR328, FR317, FR326, FR327, FR319, FR318, FR323, FR322, FR324, FR325, FR320, FR321 |
| Corollary 3.3: Finite Basis for the Current Positive Landscape                           | FR3, FR29                                                                                 |
| Corollary 6.15: No Exact Classifier on the Full Binary Pairwise Domain                   | FR350                                                                                     |
| Corollary 6.20: Least Exact Closure-Invariant Classifier                                 | FR358                                                                                     |
| Corollary 6.13: No Correct Tractability Classifier                                       | FR197, FR199, FR193, FR198                                                                |
| Corollary 6.17: Orbit-Gap Completeness on Closure-Closed Domains                         | FR210, FR212, FR213, FR211                                                                |

| <b>Paper claim</b>                                                                             | <b>Lean handle</b>                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corollary 2.22: Compute-Cost Version: Canonical Payload and Search Tasks                       | FR260, FR259, FR262, FR261                                                                                                                                                                                                     |
| Corollary 2.30: Predicate Transfer                                                             | FR341                                                                                                                                                                                                                          |
| Corollary 2.46: Randomized-Output Guarantee Semantics Inherit the Closure-Orbit Consequences   | FR295, FR293, FR292, FR291, FR294, FR296                                                                                                                                                                                       |
| Corollary 2.45: Randomized-Output Guarantee Semantics Transfer                                 | FR293, FR292, FR291                                                                                                                                                                                                            |
| Corollary 5.4: Realizability Alone Cannot Be the Frontier                                      | FR7, FR8                                                                                                                                                                                                                       |
| Corollary 6.12: Finite Structural Admissibility Packages                                       | FR194                                                                                                                                                                                                                          |
| Corollary 2.33: Arbitrary Exact Output Semantics Transfer                                      | FR231, FR230, FR229                                                                                                                                                                                                            |
| Corollary 2.3: Relevance Is Failure of Erased Sufficiency                                      | FR49, FR50                                                                                                                                                                                                                     |
| Corollary 2.36: Relevance Is Erased Failure of Refinement                                      | FR343                                                                                                                                                                                                                          |
| Corollary 2.31: Nonempty Set-Valued Payload Transfer                                           | FR225, FR224, FR223                                                                                                                                                                                                            |
| Corollary 2.50: Arbitrarily Small Uniform Perturbations Can Flip Relevance                     | FR264                                                                                                                                                                                                                          |
| Corollary 2.51: Arbitrarily Small Uniform Perturbations Can Flip Sufficiency                   | FR267                                                                                                                                                                                                                          |
| Corollary 2.7: Structural Rank Lower-Bounds Any Sufficient Description                         | FR354                                                                                                                                                                                                                          |
| Corollary 2.47: Statistical Guarantee Semantics Inherit the Closure-Orbit Consequences         | FR279, FR278, FR277, FR282, FR281, FR280, FR295, FR286, FR298, FR288, FR270, FR269, FR268, FR293, FR292, FR291, FR294, FR296, FR273, FR272, FR271, FR290, FR284, FR283, FR285, FR287, FR276, FR275, FR274, FR289, FR297, FR299 |
| Corollary 2.44: Statistical Guarantee Semantics Transfer                                       | FR279, FR278, FR277, FR282, FR281, FR280, FR270, FR269, FR268, FR273, FR272, FR271, FR276, FR275, FR274                                                                                                                        |
| Corollary 2.4: Certification Statistics Factor Through the Quotient Relation                   | FR41, FR43, FR44, FR42                                                                                                                                                                                                         |
| Corollary 2.2: Sufficiency Is Relation Refinement                                              | FR48                                                                                                                                                                                                                           |
| Corollary 2.6: Sufficient Sets Form a Principal Filter                                         | FR353                                                                                                                                                                                                                          |
| Corollary 2.32: Arbitrary Set-Valued Payload Transfer via Failure Tokens                       | FR228, FR227, FR226                                                                                                                                                                                                            |
| Corollary 2.24: Compute-Cost Version: Representation-Relative Output Objects                   | FR328, FR326, FR327, FR334, FR333, FR335, FR336, FR330, FR329, FR331, FR332                                                                                                                                                    |
| Corollary 2.38: Universal Scope Over Rigorously Specified Problems                             | FR351                                                                                                                                                                                                                          |
| Proposition 2.25: The Finite Structural Class Is Nonempty                                      | FR200                                                                                                                                                                                                                          |
| Proposition 2.11: Positive Affine Utility Reparameterizations Are Invisible                    | FR13, FR16, FR15, FR14                                                                                                                                                                                                         |
| Proposition 2.49: Approximate Relevance and Sufficiency Claims Need Explicit Stability Control | FR266, FR263                                                                                                                                                                                                                   |
| Proposition 6.1: Binary Pairwise Symmetry Dichotomy                                            | FR120, FR118, FR119                                                                                                                                                                                                            |
| Proposition 2.27: Bounded-Pattern Predicates Stabilize Above a Finite Action Bound             | FR215, FR216, FR214                                                                                                                                                                                                            |
| Proposition 2.39: Canonical Exact Relevance Profile                                            | FR302, FR303                                                                                                                                                                                                                   |
| Proposition 2.18: Closure Operations Preserve Exact Certification                              | FR57, FR17, FR18, FR60, FR56, FR59, FR84, FR16, FR55, FR58, FR83, FR15, FR19, FR20, FR85                                                                                                                                       |



| Paper claim                                                                                     | Lean handle                                                                                      |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Proposition 6.18: Closure-Invariant Predicates Are Exactly the Fixed Points of Orbit Saturation | FR355                                                                                            |
| Proposition 2.42: Countable Exact Specifications Admit Countable Boolean Presentations          | FR312, FR313, FR311                                                                              |
| Proposition 6.2: Decision-Relevant Binary Pairwise Dichotomy                                    | FR121, FR123, FR124, FR122                                                                       |
| Proposition 4.2: Degenerate Positive Cases                                                      | FR1, FR6                                                                                         |
| Proposition 2.1: Exact Certification Depends Only on the Decision Quotient Relation             | FR32, FR33, FR31, FR30, FR37, FR51, FR35, FR34                                                   |
| Proposition 2.26: Bounded Distinct Action Profiles Compress to Bounded Actions                  | FR206, FR207, FR205, FR209, FR208                                                                |
| Proposition 2.12: Duplicate Actions and Duplicate States Preserve Certification                 | FR71, FR68, FR54, FR70, FR56, FR69, FR55, FR73, FR72                                             |
| Proposition 2.10: Explicit Enumeration Is Parameter-Dependent, Not Structural                   | FR66, FR67                                                                                       |
| Proposition 2.40: Every Exact Specification Admits a Coordinate Presentation                    | FR307, FR308                                                                                     |
| Proposition 2.34: Exactness Means Exact Agreement With Validity                                 | FR346                                                                                            |
| Proposition 3.4: Explicit Primitive Basis                                                       | FR29                                                                                             |
| Proposition 3.2: Explicit Inventory                                                             | FR26, FR25, FR3, FR28, FR27                                                                      |
| Proposition 2.41: Finite Exact Specifications Admit Coordinate Presentations                    | FR304, FR305, FR306                                                                              |
| Proposition 2.43: Finite Exact Specifications Admit Low-Dimensional Boolean Presentations       | FR314, FR315                                                                                     |
| Proposition 3.5: All Independent Core Mechanisms Are Nondegenerate                              | FR108, FR106, FR112, FR101, FR105, FR113                                                         |
| Proposition 6.14: Full Binary Pairwise Domain Is Closure-Closed                                 | FR348, FR349                                                                                     |
| Proposition 2.52: Global Approximation Stability Under Uniform Strict Gaps                      | FR300, FR301, FR295, FR298, FR288, FR293, FR292, FR291, FR294, FR296, FR290, FR289, FR297, FR299 |
| Proposition 3.9: Current Positive Interfaces Factor Through Role Classification                 | FR4                                                                                              |
| Proposition 2.15: Invariance Alone Is Too Weak                                                  | FR22, FR21, FR23                                                                                 |
| Proposition 3.6: Weak Tree Ordering Does Not Imply Width One                                    | FR115, FR114                                                                                     |
| Proposition 4.1: Lifted Cases Introduce No New Primitive                                        | FR5                                                                                              |
| Proposition 6.11: Logical Dependency of the No-Go                                               | FR347                                                                                            |
| Proposition 2.5: Minimal Sufficient Sets Are Canonical on Product Spaces                        | FR352                                                                                            |
| Proposition 2.17: Unrestricted Predicates Trivialize Exact Characterization                     | FR145                                                                                            |
| Proposition 6.16: Orbit-Gap Completeness for Exact Classification                               | FR203, FR201, FR202, FR204                                                                       |
| Proposition 6.3: Orbit-Gap Template                                                             | FR185                                                                                            |
| Proposition 3.8: Parent-Tree Structure Gives Width-One Decompositions                           | FR110, FR100, FR111                                                                              |
| Proposition 2.28: Deterministic Payload Transfer                                                | FR219, FR220, FR222, FR218, FR221, FR217                                                         |
| Proposition 2.14: Quotient Size Is Unbounded Under Realizability                                | FR24, FR38                                                                                       |
| Proposition 5.3: The Realized Quotient Is Exactly the Label Range                               | FR39, FR40                                                                                       |
| Proposition 2.13: Relabeling Invariance Is Forced by Exact Certification                        | FR17, FR18, FR19, FR20                                                                           |

| <b>Paper claim</b>                                                                            | <b>Lean handle</b>                                     |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Proposition 2.53: Exact Semantic Claims Depend Only on Admissible-Output Equivalence          | FR236, FR235                                           |
| Proposition 2.55: Semantically Extensional Claims Factor Through the Exact-Semantics Quotient | FR242, FR241, FR244, FR243                             |
| Proposition 5.2: Every Equivalence Relation Is Optimizer-Realizable                           | FR47, FR46, FR45                                       |
| Proposition 2.16: Independent Summary Frameworks Converge on the Same Structural Core         | FR178, FR181, FR182, FR180, FR179                      |
| Proposition 2.35: Sufficiency Is Relation Refinement                                          | FR342                                                  |
| Proposition 3.7: Cyclic Dependencies Recover Hardness                                         | FR110, FR100, FR111, FR115                             |
| Proposition 2.8: Zero-Distortion Summaries Refine the Optimizer Quotient                      | FR177, FR176                                           |
| Theorem 6.9: Finite-Structural Collapse Across the Four Obstruction Families                  | FR190                                                  |
| Theorem 6.10: Closure-Sound Package No-Go                                                     | FR193                                                  |
| Definition 2.20: Finite Structural Predicate                                                  | FR197, FR199, FR198, FR249, FR248, FR245, FR246, FR247 |
| Theorem 3.1: Explicit Partition of the Current Positive Landscape                             | FR26, FR25, FR1, FR2, FR28, FR27                       |
| Theorem 6.5: Dominant-Pair Finite-Structural No-Go                                            | FR186                                                  |
| Theorem 6.19: Exact Classification Equals Hull Separation                                     | FR357, FR356                                           |
| Theorem 2.37: Exact Semantics Quotient Universality                                           | FR345                                                  |
| Theorem 6.7: Ghost-Action Finite-Structural No-Go                                             | FR188                                                  |
| Theorem 5.1: Every Labeling Kernel Is Optimizer-Realizable                                    | FR7, FR8                                               |
| Theorem 6.6: Margin-Masking Finite-Structural No-Go                                           | FR187                                                  |
| Theorem 6.8: Offset Finite-Structural No-Go                                                   | FR189                                                  |
| Theorem 2.21: Compute-Cost Version: Optimizer Computation                                     | FR248                                                  |

*Auto summary: mapped 89/89 (full=89, derived=0, unmapped=0).*